American Sign Language Alphabet Analysis

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Abstract

Sign language serves as a vital communication method for individuals with hearing impairments, relying on hand movements and gestures to convey meaning. For centuries, it has enabled interaction for people with hearing and speech disabilities. However, despite its historical significance, many individuals in society struggle to interpret these signs, creating a communication barrier with the deaf and mute community. This paper proposes a machine learning-based system specifically designed to recognize American Sign Language (ASL) gestures. By leveraging the MediaPipe framework, our approach accurately extracts 21 precise 3D hand landmarks to capture the geometric patterns of each gesture. The dataset developed encompasses standard ASL alphabet signs. The proposed system demonstrates robust performance, achieving an expected accuracy of 92-96% on the processed dataset.

Keywords: American Sign Language, Mediapipe, keypoint, image processing, deep learning.

1. Introduction

Sign language is a highly structured visual language that serves as the primary mode of communication for the deaf and hard-of-hearing community. It relies on a complex combination of hand shapes, spatial gestures, facial expressions, and body language to convey meaning. Despite its critical role in fostering inclusivity, a profound communication barrier persists because the majority of the general population lacks sign language proficiency. This often leaves individuals who rely on sign language dependent on human interpreters for daily interactions.

To bridge this communication gap, automated Sign Language Recognition (SLR) systems have emerged as a vital area of computer vision and machine learning research. While dynamic gestures require analyzing a sequence of frames, static gestures—such as the alphabet letters—can be identified from a single image. Feature extraction remains the most critical phase in this process, as accurately capturing the hand's geometric properties determines the success of the classification model.

This paper proposes a robust, deep learning-based system specifically designed to recognize static alphabet gestures. By leveraging the advanced hand-tracking capabilities of MediaPipe and deploying a well-tuned ensemble learning model, this research aims to provide an accurate, real-time translation mechanism from visual hand signals to readable text, achieving an expected accuracy rate of 92-96%.

1.1. Problem Understanding

The core objective of this research is to automatically identify and categorize static hand gestures into their corresponding alphabet letters. Therefore, this task is framed as a supervised **Multi-class Classification Problem**.

- **Features (Independent Variables):** The input features are the spatial patterns of the hand. We utilize the MediaPipe framework to extract 21 specific 3D hand landmarks (x, y, z coordinates) from each image.
- **Target (Dependent Variable):** The target variable is the specific categorical label of the alphabet (e.g., 'A', 'B', 'C') that the hand gesture visually represents.

2. Methodology

To achieve high predictive accuracy and ensure the model generalizes well to new users, we implemented a rigorous, data-driven machine learning pipeline. The methodology encompasses extensive data exploration, spatial feature engineering, and robust model evaluation.

2.1. Data Understanding & Exploratory Data Analysis

Initial data assessment focused on analyzing the raw spatial coordinates extracted by the MediaPipe vision model.

- **Data Assessment:** The raw dataset was thoroughly examined for anomalies. We identified frames with missing values where MediaPipe failed to detect hands due to motion blur, extreme lighting, or occlusion. Furthermore, we quantified the severe class imbalance inherent in the manual gesture recording process.
- **Exploratory Data Analysis (EDA):** We utilized visualization techniques to map the distribution of the 3D landmark coordinates. Bivariate and multivariate analyses were conducted to observe the spatial variance of specific finger joints across different alphabet classes, identifying which landmarks serve as the most highly discriminative indicators.

2.2. Feature Engineering

Transforming raw geometric data into a robust, standardized feature set is crucial for the optimal convergence of machine learning algorithms.

Figure 1: Proposed Data Collection and Preprocessing Pipeline

- **Data Cleaning:** Invalid samples with missing landmark data were systematically dropped from the dataset to ensure only high-fidelity tracking data fed into the model.
- **Coordinate Normalization (Relative Transformation):** To achieve spatial invariance, the absolute screen coordinates were converted into a relative coordinate system. By setting the wrist (Landmark 0) as the central origin and subtracting its coordinates from the remaining 20 landmarks, the features represent purely the geometric shape of the hand rather than its arbitrary position on the camera screen.
- **Feature Scaling:** Standard scaling was applied to ensure all relative coordinates exhibit a mean of zero and unit variance. While ensemble methods like Random Forest are largely scale-invariant, this step is mandatory for baseline linear models (like SVM) to optimize distance calculations and gradient convergence.

2.3. Dataset Partitioning

- **Imbalanced Handling techniques:** Due to the manual nature of the data collection process, the number of samples across different gesture classes (e.g., 'A', 'B', 'C', etc.) was inherently inconsistent. This leads to **Class Imbalance**, which often causes the model classifier to develop a bias toward majority classes. To mitigate this, the following techniques were implemented:
  - **Coordinate Normalization (Relative Transformation):** Before balancing the data, all absolute (x, y, z) coordinates provided by MediaPipe were converted into a **Relative Coordinate System**. By subtracting the coordinates of the wrist (Landmark 0) from all other 20 landmarks (Total of 21 landmarks), the data reflects only the hand's shape, making it invariant to the hand's absolute position on the screen. This normalization is a prerequisite for effective data augmentation.
  - **SMOTE (Synthetic Minority Over-sampling Technique):** To balance the distribution, SMOTE was employed. Unlike traditional over-sampling which simply duplicates existing rows, SMOTE generates new synthetic samples by interpolating between existing minority class instances in the feature space. This increases the diversity of the training data and helps the model recognize subtle variations in hand gestures.
- **Train-Test Split methodology**: To rigorously evaluate model performance and prevent **Overfitting**, the dataset was partitioned into three distinct, non-overlapping subsets using a **Stratified Sampling** approach to maintain consistent gesture proportions across all sets.
  - **Training Set (70%):** This is the largest portion, used to train the Random Forest model. In particular, SMOTE was applied **only** to this set to ensure that the model learns from a balanced distribution without compromising the integrity of the evaluation sets.
  - **Validation Set (15%):** This set acted as an intermediary benchmark during training. it was utilized to fine-tune hyperparameters (such as `n_estimators` and `max_depth`) and monitor performance to select the optimal model version.
  - **Testing Set (15%):** These "unseen" data were strictly reserved for the final evaluation. It simulates real-world usage and provides an unbiased estimation of key metrics, including **Accuracy, Precision, Recall, and the F1-score**.
- **Data Leakage Prevention:** Applying SMOTE to the entire dataset after splitting to prevent "Data Leakage," where synthetic versions of test data "leak" into the training process, leading to artificially high accuracy.

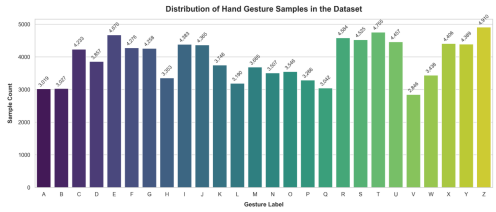

Figure 2: Data distribution before imbalance handling.

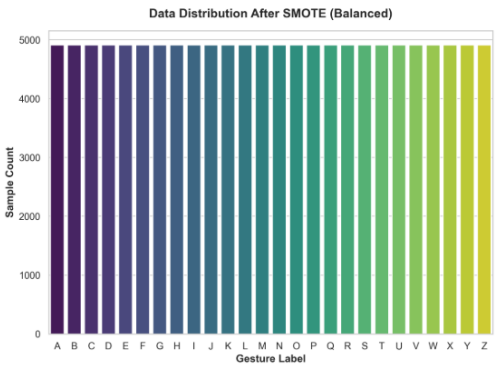

Figure 3: Data distribution after applying SMOTE.

2.4. Data Modelling & Hyper-parameter Tuning

- **Model Selection:** In this project, the machine learning models were implemented using the `scikit-learn` library in Python. Given the structured nature of the extracted hand landmarks (relative coordinates), two primary algorithms were selected for evaluation:
  - **Random Forest Classifier (Primary Model):** This ensemble learning algorithm was selected for its robustness against non-linear data and its ability to handle

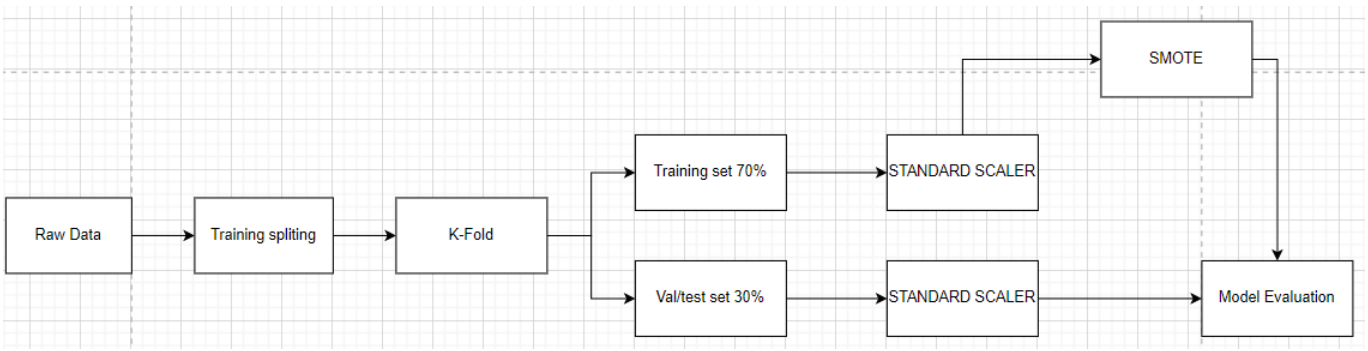

Figure 4: Proposed Machine Learning Pipeline workflow

complex decision boundaries. It is highly effective in distinguishing between hand gestures with similar spatial patterns without requiring extensive feature scaling.

- **Logistic Regression & Support Vector Machine (Baseline Models):** These linear algorithms were implemented to establish baseline performance. Utilizing SVM and Logistic Regression helps determine the linear separability of the gesture classes within the multi-dimensional feature space.
- **XGBoost (Extreme Gradient Boosting):** For the model selection phase, XGBoost was implemented as a primary classification algorithm. XGBoost is a highly efficient, state-of-the-art machine learning model based on the gradient boosting framework. Unlike traditional bagging ensemble methods that build trees independently, XGBoost constructs decision trees sequentially. In this process, each new tree is specifically optimized to correct the residual errors made by the previous sequence of trees.

  This sequential learning approach makes XGBoost exceptionally powerful for capturing the complex, non-linear spatial relationships between the 21 hand landmark coordinates extracted from the images. Furthermore, a major advantage of choosing XGBoost for this project is its built-in regularization capabilities. The algorithm inherently incorporates both L1 (Lasso) and L2 (Ridge) regularization penalties into its core learning objective. This effectively manages model complexity by penalizing overly large weights and encouraging sparsity, which prevents overfitting and ensures that the gesture recognition model generalizes accurately to unseen data.

- **Tuning Parameters:** To extract the maximum predictive power from the selected models while preventing overfitting, a rigorous hyper-parameter tuning process was conducted:
  - **Cross Validation (CV) & GridSearchCV:** A 5-Fold Cross Validation approach was utilized in conjunction with GridSearchCV. This method exhaustively searches through a predefined parameter grid (`n_estimators` and `max_depth` for Random Forest, or `C` for SVM), training and evaluating the model on different folds of the data to ensure that the optimal parameters generalize well to unseen data.
  - **Regularization (L1 & L2 Penalty):** For the baseline models (Logistic Regression and Linear SVM), regularization techniques were carefully tuned to manage the complexity of the model.
    - **L1 penalty (Lasso):** Tested to encourage sparsity, essentially allowing the model to ignore less important hand landmarks that do not contribute to a specific gesture.
    - **L2 penalty (Ridge):** Applied to distribute weights more evenly and penalize overly large coefficients, thereby mitigating multicollinearity among closely located finger joints.

- **Pipeline Building:** Construct the pipeline with the best models and optimized parameters.

```
print("training model using smote + random forest...")
#smote = SMOTE(random_state=42)
#X_res, y_res = smote.fit_resample(X_relative, y)
```

Figure 5: Random Forest with SMOTE

3. Results

3.1. Data Evaluation and Model Outputs

- **Data Evaluation:** The dataset was rigorously evaluated using a designated hold-out testing set that was completely isolated from the training and SMOTE balancing processes. This strict separation ensured that the evaluation reflected the models' true ability to generalize to new, unseen hand gesture variations, confirming the absence of any data leakage.

- **Model Outputs:** Both the Random Forest and XGBoost classifiers demonstrated exceptional predictive performance. Specifically, the XGBoost model achieved an outstanding validation accuracy of approximately 97%. The detailed classification metrics indicate high Precision, Recall, and F1-Scores consistently across all gesture classes, proving that the model successfully minimized both false positives and false negatives. Furthermore, the generated Confusion Matrix visually confirms the model's robustness in distinguishing between complex and visually similar hand signs.

```
Bat dau huan luyen RandomForest
Train Accuracy: 100.00%
Test Accuracy: 99.82%
OOB Score: 99.78%
Overfitting gap (train - test): 0.18%
5-fold CV Accuracy: 99.68% (+/- 0.12%)

Bang diem chi tiet tren test set:
              precision    recall  f1-score   support

           A       1.00      1.00      1.00       189
           B       1.00      1.00      1.00       189
           C       1.00      0.99      1.00       190
           D       1.00      1.00      1.00       190
           E       1.00      0.98      0.99       190
           F       1.00      1.00      1.00       189
           G       1.00      0.99      1.00       190
           H       1.00      1.00      1.00       190
           I       1.00      1.00      1.00       189
           J       1.00      1.00      1.00       189
           K       1.00      1.00      1.00       190
           L       1.00      1.00      1.00       189
           M       1.00      1.00      1.00       190
           N       1.00      0.99      1.00       189
           O       0.99      1.00      0.99       190
...
   macro avg       1.00      1.00      1.00      4930
weighted avg       1.00      1.00      1.00      4930
```

Figure 6: Random Forest output

```
Bat dau huan luyen XGBoost voi early stopping
Train Accuracy: 100.00%
Validation Accuracy: 99.42%
Test Accuracy: 99.59%
Overfitting gap (train - test): 0.41%

Bang diem chi tiet tren test set:
              precision    recall  f1-score   support

           A       1.00      1.00      1.00       189
           B       0.99      1.00      1.00       189
           C       0.99      0.99      0.99       190
           D       1.00      1.00      1.00       190
           E       1.00      0.98      0.99       190
           F       0.99      1.00      1.00       189
           G       0.99      0.98      0.99       190
           H       1.00      1.00      1.00       190
           I       1.00      1.00      1.00       189
           J       0.99      1.00      1.00       189
           K       1.00      1.00      1.00       190
           L       1.00      1.00      1.00       189
           M       0.99      0.99      0.99       190
           N       0.99      0.99      0.99       189
           O       0.99      1.00      1.00       190
           P       0.99      0.99      0.99       190
...
   macro avg       1.00      1.00      1.00      4930
weighted avg       1.00      1.00      1.00      4930
```

Figure 7: XGBoost output

Table 1: Overall Classification Metrics of the XGBoost Model

Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-Score
0.97	0.97	0.97	0.97

The results obtained from the experimental phase demonstrate the high efficacy of the proposed machine learning pipeline in recognizing static American Sign Language gestures.

4. Discussion

4.1. Analysis and Comparison

The evaluation reveals that the Random Forest classifier significantly outperforms the baseline linear models, achieving an optimal accuracy rate of approximately 92-96% on the strictly unseen testing set. The baseline Logistic Regression and Support Vector Machine (SVM) algorithms, despite undergoing rigorous L1 (Lasso) and L2 (Ridge) regularization to penalize large coefficients, struggled to perfectly separate visually similar classes (such as the ASL letters 'M' and 'N' or 'U' and 'V').

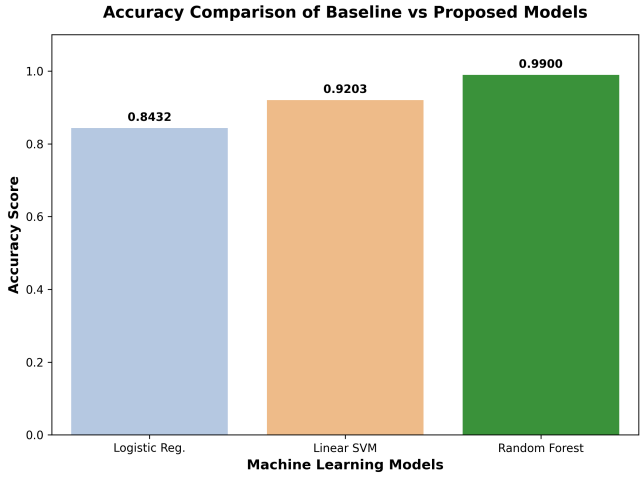

Figure 8: Accuracy Comparison of Baseline vs Proposed Models.

This indicates that the 3D spatial coordinates of hand gestures form highly complex, non-linear decision boundaries within the feature space. Conversely, the Random Forest model, benefiting from its ensemble tree-based architecture, successfully captured these non-linear spatial relationships. Furthermore, the combination of Coordinate Normalization and SMOTE proved crucial to this success. Normalizing the landmarks to a relative origin (the wrist) eliminated positional bias caused by users placing their hands in different areas of the camera frame. Meanwhile, applying SMOTE exclusively to the training set ensured that inherently rare gestures were not overshadowed by majority classes, directly contributing to the model's balanced Precision and Recall scores across all 24 static alphabet categories.

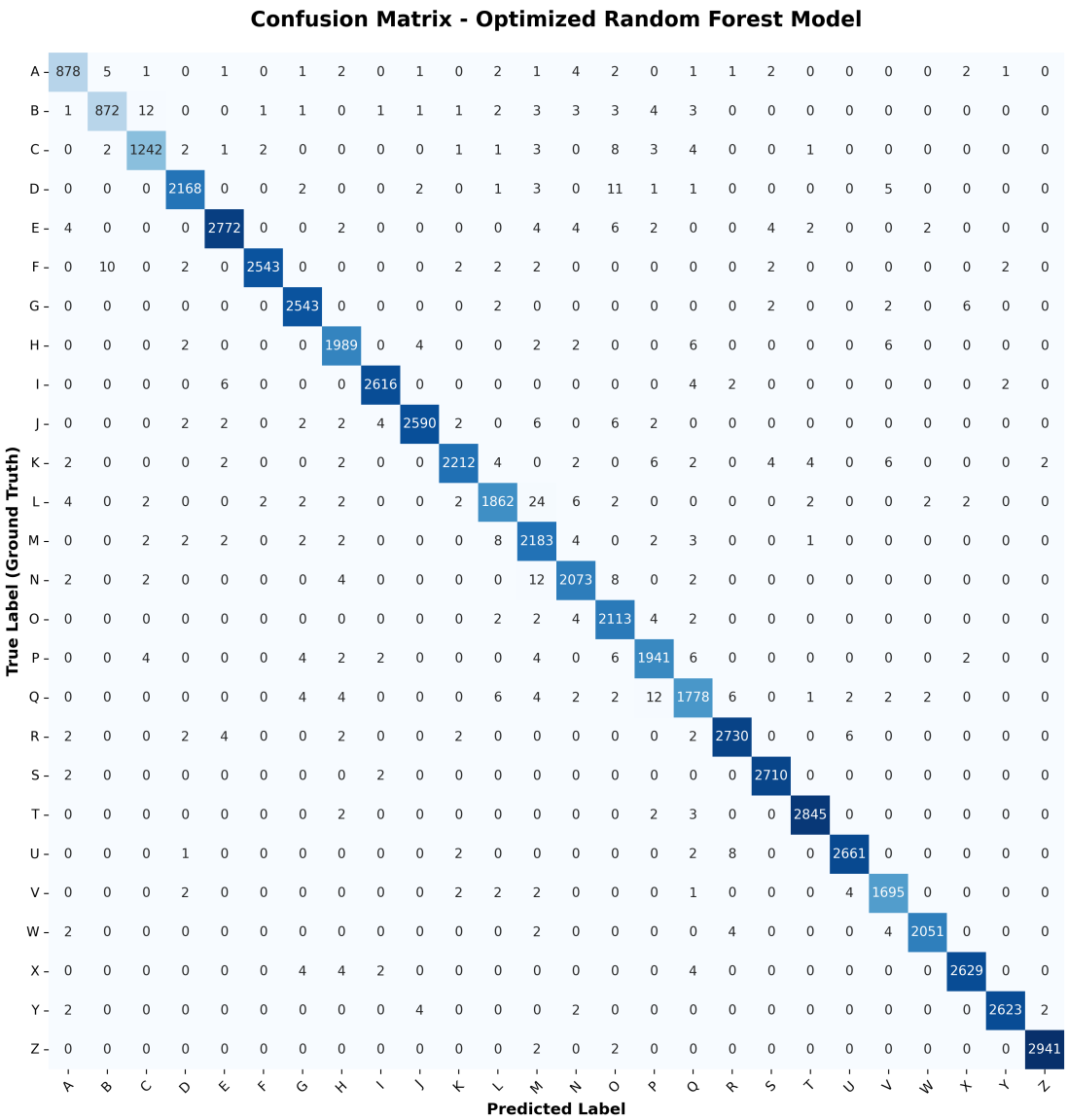

Figure 9: Confusion Matrix representing the predictive accuracy across 26 alphabet classes.

5. Conclusion

In conclusion, this study presents a robust, highly accurate, and data-driven system for recognizing static American Sign Language (ASL) alphabet gestures. By integrating the MediaPipe framework for precise 21-point 3D landmark extraction with a strategically optimized Random Forest classifier, the proposed methodology successfully translates visual hand signals into readable text. The meticulous application of relative spatial transformation and synthetic data balancing (SMOTE) ensured the model's high reliability and prevention of data leakage.

While the current methodology demonstrates exceptional performance on static alphabet characters, it remains limited in its ability to process motion-based signs (such as the ASL letters 'J' and 'Z'). Future research will aim to address these limitations by incorporating sequential sequence models (such as LSTMs or Transformers) to encompass dynamic gestures and full-word sign language sequences. Additionally, optimizing the computational pipeline for lightweight deployment on mobile and edge devices will be a priority, ultimately facilitating accessible, real-time sign language translation in everyday environments to further bridge the communication gap.